

Performance Testing

ComicCraft - AI Comic Story Creator using Gemini Models
Madras Institute of Technology | Department of Computer Technology (CT) | III Year, V Semester
Project period: 1 Oct 2026 - 2 Oct 2026

Environment

Apple M5, 16 GB unified memory, PyTorch MPS, float16, Stable Diffusion v1.5, 25 inference steps, 512x512, attention slicing.

Measurements

Test	Result
Time per panel image after warm-up	about 12.4 s (measured)
Estimated 5 panels	about 62 s (5 x 12.4 s)
End-to-end run (form submit to preview, 5 panels + PDF, measured)	104 s (HTTP 200, 5-page PDF produced)
Gemini calls per comic	2 (outline + story)
Story generation	runs in parallel with images, adds no extra wait
First request without warm-up	slower, model load time added; avoided by start-up warm-up

Optimisations applied

- fp16 on MPS/CUDA instead of fp32
- Story generation overlaps image generation
- Pipeline loaded once at server start
- Quality settings (25 steps, 512 px, safety checker) were kept; cutting them was tested and reverted

Functional checks

Check	Result
GET / returns 200	Pass
Outline returns 5 panels	Pass
Story parses into panels	Pass
Images saved	Pass
PDF created	Pass
Gemini quota error shown clearly	Pass (HTTP 500 with message)

Evidence



Preview page after the 104 s end-to-end run

Automated tests

Suite	Result
tests/test_app.py (pytest, FastAPI TestClient, mocked Gemini and image calls)	10 passed
Covers	layout parsing, PDF writing, home page, /generate, JSON API, validation, quota error path, /export-success, /test-image

Next

Run the pytest suite in CI on every push.